

Feedback — Lecture 4 Quiz

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You submitted this quiz on **Sat 18 Jul 2015 12:12 PM PDT**. You got a score of **9.00** out of **9.00**.

Question 1

The following expression is true when rnatype is 'ncRNA' and length is at least 200, or rnatype is 'ncRNA' and length is 22:

(rnatype is 'ncRNA' and length>=200) or (rnatype is 'ncRNA' and length==22)

What Boolean expression below represents a negation of the above expression?

Your Answer	Score	Explanation
☐ (rnatype is not 'ncRNA' and length		
☐ (rnatype is not 'ncRNA' and length		
☐ rnatype is not 'ncRNA' and (length		
☒ rnatype is not 'ncRNA' or (length	✓ 1.00	
Total	1.00 / 1.00	

Question 2

For what values of the variable fold would the following code print 'condition B'?

```
if fold > 2 : print('condition A')
```

```
elif fold>100: print('condition B')
```

```
if fold> 2 or fold<2 : print('condition A')
```

```
else : print('condition B')
```

Your Answer	Score	Explanation
☒ if fold is 2	✓ 1.00	
☐ if fold is bigger than 100		
☐ if fold is bigger than 100 or fold is 2		

☐ never

Total

1.00 / 1.00

Question 3

How many times will Python execute the code inside the following while loop?

```
i=1
```

```
while i< 2048 :
```

```
i=2*i
```

Your Answer	Score	Explanation
☐ 1024		
☒ 11	✓ 1.00	
☐ 2048		
☐ 12		
Total	1.00 / 1.00	

Question 4

What sequence of numbers does the range(1,-23,-3) expression evaluate to?

Your Answer	Score	Explanation
☒ 1, -2, -5, -8, -11, -14, -17, -20	✓ 1.00	
☐ -23, -21, -19, -17, -15, -13, -11, -9, -7, -5, -3, -1, 1		
☐ 1, -1, -3, -5, -7, -9, -11, -13, -15, -17, -19, -21, -23		
☐ -22, -19, -16, -13, -10, -7, -4, -1		
Total	1.00 / 1.00	

Question 5

A substring in programming represents all characters from a string, between two specified indices. Given a variable string called `seq`, a student writes the following program that will generate all nonempty substrings of `seq`:

```
for i in range(len(seq)) :   # line 1
    for j in range(i) :      # line 2
        print(seq[j:i])     # line 3
```

Which of the following changes make the above program correct?

- A. Program is correct as it is.
- B. Change line 1 to: `for i in range(len(seq)+1) :`
- C. Change line 3 to: `print(seq[j:i+1])`
- D. Change line 2 to: `for j in range(i+1) :`

Your Answer	Score	Explanation
☒ Only B	1.00	
☐ Only A		
☐ Only C		
☐ Only D		
Total	1.00 / 1.00	

Question 6

While and for loops are equivalent: whatever you can do with one you can do with the other. Given the for loop written by the student in the previous problem, which of the following while loops are equivalent to it:

- A.
- `i=0`
- `while i<len(seq) :`
- `j=0`

```
while(j<i) :  
    print(seq[j:i])
```

B. i=1 while i<len(seq) :

```
j=1  
while(j<i) :  
    print(seq[j:i])  
    j=j+1  
i=i+1
```

C.

```
i=0  
while i<len(seq) :  
    j=0  
    while(j<i) :  
        print(seq[j:i])  
        j+=1  
    i+=1
```

D.

```
i=0  
while i<len(seq)+1 :  
    j=0  
    while(j<i+1) :  
        print(seq[j:i])  
        j=j+1  
    i=i+1
```

E.

```
i=1  
while i<len(seq)+1 :  
    j=1  
    while(j<i+1) :  
        print(seq[j:i])
```

F.

```
i=0
```

```
while i<len(seq) :  
    j=i  
    while(j>0) :  
        print(seq[j:i])  
        j=j+1  
    i=i+1
```

Your Answer	Score	Explanation
☒ C only	✓ 1.00	
☐ A, B, and D only		
☐ C, and D only		
☐ C, D, and E only		
Total	1.00 / 1.00	

Question 7

Study the following two Python code fragments:

Version 1.

```
d = {}  
result = False  
for x in mylist:  
    if x in d:  
        result=True  
        break  
d[x] = True
```

Version 2.

```
d = {}  
result = False  
for x in mylist:  
    if not x in d:  
        d[x]=True  
        continue
```

result = True

Both versions should determine if there is any element that appears more than once in the list mylist. If there is such an element then the variable result should be True, otherwise it should be False. For instance, if mylist=[1,2,3,3] the result variable should be True. Which of the following statements is True for any value of the list mylist after the execution of both versions of code?

Your Answer	Score	Explanation
☒ The value of the result variable is the same, but the variable d is different.	✓ 1.00	
☐ Neither Version 1 or Version 2 are computing the value of the result variable correctly.		
☐ Version 1 is not computing the result variable correctly.		
☐ Version 2 is not computing the result variable correctly.		
Total	1.00 / 1.00	

Question 8

Study the following if statement:

```
if x>10 or x<=-10: print('big')
elif x>1000000: print('very big')
elif x<-1000000: print('very big')
else : print('small')
```

For what values of x will the above code print 'very big'?

Your Answer	Score	Explanation
☒ For no value	✓ 1.00	
☐ For x > 1000000		
☐ For x		
☐ For x > 1000000 or x		
☐ For x = -10		
Total	1.00 / 1.00	

Question 9

What will be the value of the variable i after the following code is executed:

```
i = 1
```

```
while i < 100:
```

```
    if i%2 == 0 : break
```

```
    i += 1
```

```
else:
```

```
    i=1000
```

Your Answer	Score	Explanation
☐ 99		
☒ 2	✓ 1.00	
☐ 1		
☐ 100		
☐ 98		
☐ 1000		
Total	1.00 / 1.00	